

Multiple Choice

1. **Answer:** C

Options: A) K⁺; B) Ca₂⁺; C) Sc₃⁺; D) Rb⁺

Note: Correct option: C - Sc₃⁺

2. **Answer:** D

Options: A) Metallic element; B) Sulfide; C) Fluoride; D) Oxide

Note: Correct option: D - Oxide

3. **Answer:** D

Options: A) The orbitals are degenerate.; B) The set of orbitals has a tetrahedral geometry.; C) These orbitals are constructed from a linear combination of atomic orbitals.; D) Each hybrid orbital may hold four electrons.

Note: Correct option: D - Each hybrid orbital may hold four electrons.

4. **Answer:** A

Options: A) Cu; B) Gd; C) Eu; D) Am

Note: Correct option: A - Cu

5. **Answer:** A

Options: A) Blackbody radiation curves; B) Emission spectra of diatomic molecules; C) Electron diffraction patterns; D) Temperature dependence of reaction rates

Note: Correct option: A - Blackbody radiation curves

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'trigonal pyramidal'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'most organic functional groups absorb in a characteristic region of the IR spectrum'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'NH₂'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Neutrons. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 9 lines. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key